



ME 300: Engineering Thermodynamics

Example: Diesel Cycle

Examples

An engine runs an air-standard diesel cycle with a compression ratio of 18, a cut-off ratio of 2.37 using an ideal cycle. The displacement of the engine is 0.002 m³ (2 L). If the intake air is $T=300$ K and $P_1=0.1$ MPa, calculate the following ($R=287$ J/kg-K, $c_v=714$ J/kg-K):

- Work done by the cycle
- Heat added during combustion
- Efficiency of the cycle

Given: $v_1/v_2 = 18$

$$v_3/v_2 = 2.37$$

$$v_1 - v_2 = 0.002 \text{ m}^3$$

$$T_1 = 300 \text{ K}$$

$$P_1 = 0.1 \text{ MPa}$$

Find: $\left. \begin{matrix} \bar{W}_{\text{net}} \\ Q_{\text{in}} \end{matrix} \right\} \eta$

Assume: air = ideal gas

$c_v = \text{const.}$

quasi-steady

reversible

$$\Delta KE = \Delta PE = 0$$

Strategy:

$$\bar{W}_{\text{net}} = \bar{W}_1 + \bar{W}_2 + \bar{W}_3 + \bar{W}_4$$

$$\Rightarrow \bar{W} = \int P dv$$

$$\underline{\underline{\Delta U}} = Q - \bar{W} = M c_v \Delta T$$

$$Q_{\text{in}} = {}_2Q_3$$

$$\eta = \frac{\bar{W}_{\text{net}}}{Q_{\text{in}}}$$

State 1 and 2

State 1: $T_1 = 300 \text{ K}$

$$P_1 = 0.1 \text{ MPa}$$

$$V_1 = \underline{\quad}$$

$$P_1 V_1 = M R T_1$$

$$M = \frac{P_1 V_1}{R T_1}$$

$$= \frac{(100000)(0.00212)}{(287)(300)}$$

$$\boxed{M = 0.00246 \text{ kg}}$$

$$\frac{V_1}{V_2} = 18 \rightarrow V_1 = 18 V_2$$

$$V_1 - V_2 = 0.002 \Rightarrow 18 V_2 - V_2 = 0.002$$

$$\rightarrow V_2 = 0.00012 \text{ m}^3$$

$$V_1 = 0.00212 \text{ m}^3$$

State 2: $V_2 = 0.00012 \text{ m}^3$

$$P_2 = P_1 \left(\frac{V_1}{V_2} \right)^{\gamma} = 100000 (18)^{1.4} \rightarrow P_2 = 5719808 \text{ Pa}$$

$$T_2 = \frac{P_2 V_2}{M R} = \frac{(5719808)(0.00012)}{(0.00246)(287)} \rightarrow T_2 = 972.2 \text{ K}$$

State 3 and 4

$$P_3 = P_2 = 5719808 \text{ Pa}$$

$$V_3 = \alpha V_2 = (2.37)(0.00012) = 0.00028 \text{ m}^3$$

$$T_3 = \frac{P_3 V_3}{MR} = \frac{(5719808)(0.00028)}{(0.00246)(287)} = 2268.4 \text{ K}$$

State 4 : $P_4 = P_3 \left(\frac{V_3}{V_4} \right) = 5719808 \left(\frac{0.00028}{0.00212} \right)^{1.4} = 336149.5 \text{ Pa}$

$\uparrow$
 V_1

$$T_4 = \frac{P_4 V_4}{MR} = \frac{(336149.5)(0.00212)}{(0.00246)(287)} = 1009.4 \text{ K}$$

Work

$$1-2: \Delta \bar{U} = \cancel{Q_2}, \bar{W}_2, \bar{W}_2 = -\Delta \bar{U} = -Mc_v(T_2 - T_1) = -(0.00246)(714)(972.2 - 300) \\ = -1180.7 \text{ J}$$

$$2-3: {}_2\bar{W}_3 = \int_{v_2}^{v_3} P dv = P_2(v_3 - v_2) = 5719808(0.00028 - 0.00012) \\ = 915.17 \text{ J}$$

$$3-4: {}_3\bar{W}_4 = -\Delta \bar{U} = -Mc_v(T_4 - T_3) = -(0.00246)(714)(2268.4 - 1009.4) \\ = 2211.4 \text{ J}$$

$$\boxed{\bar{W}_{\text{net}} = 1946.4 \text{ J}}$$

Heat and efficiency

$${}_2Q_3 \rightarrow \Delta \bar{U} = {}_2Q_3 - {}_2\bar{W}_3$$

$${}_2Q_3 = \Delta \bar{U} + {}_2\bar{W}_3$$

$$= M C_v (T_3 - T_2) + {}_2\bar{W}_3$$

$$= (0.00246)(714)(2268.4 - 972.2) + 915.17$$

$$\boxed{{}_2Q_3 = 3191.9 \text{ J}}$$

$$\eta_{th} = \frac{\bar{W}_{net}}{Q_{in}} \rightarrow \boxed{\eta_{th} = 0.61}$$